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Modeling and optimization of photovoltaic thermal system under recyclic operation by response surface methodology

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ABSTRACT

Cooling of the photovoltaic modules is an important requirement to enhance their performance and lifespan. Therefore, a significant enhancement in heat transfer and electrical characteristics of photovoltaic thermal (PVT) collectors under recycling operation has become one of the attractive options. In the present study, the statistical models have been introduced by applying the response surface methodology (RSM) to examine the performance parameters viz; thermal efficiency, electrical efficiency, the temperature of PV module, thermal energy gain, and net electrical power of recycling type PVT system and correlate them with the operational parameters, including the packing factor, mass flow rate, recycle and channel depth ratio. For data formulation, a standard RSM design of 2^4 full factorials called a central composite design (CCD) has been used. Analysis of variance (ANOVA) revealed the value of the significant coefficient (R^2) closer to one, which means the superior response model fits the analytical data and the optimized values of the operational parameters for the efficient performance of the PVT system are established. ANOVA findings demonstrate that mass flow rate is the most significant input factor influencing the thermal and electrical performance of the PVT system.

1. Introduction

With rising energy demand, rising energy prices, and increased efforts to combat global warming, renewable sources of energy have taken hold in energy advertising. Energy has become an essential aspect of daily life as technologies have advanced. The economy depends on conventional fuels to meet its energy demands, but due to the diminution of fossil fuel reserves and ecological problems [1], the usage of renewable energy is becoming increasingly significant [2]. Countries are turning to nontoxic renewable energy sources for declining CO_2 and other harmful gaseous. Solar energy is eco-friendly and abundant across the world all year, thus it may be used by huge portions of society as a desirable alternative to meet energy demands, particularly for domestic applications [3]. Photovoltaic has the highest initial cost of any existing sustainable energy source, yet it is widely used due to its lower maintenance and operation expenses [4]. Other benefits of solar PV include higher efficiency and non-polluting electricity [5]. One of the major negative aspects of PV cells is their low efficiency as a result of high temperatures produced by excessive sun exposure [6]. The usage of hybrid photovoltaic thermal (PVT) systems is the most significant approach for decreasing the PV module's temperature [7]. The hybrid PVT system uses solar and wind energy as key power sources; hybrid energy systems are typically more reliable and less expensive than other systems that use a single source of energy [8].

Various approaches have been employed in numerous studies to cool the PVT system with air, water, and a mixture of air/water working as heat transfer fluids (HTF). Cooling the Photovoltaic panel by using air/water as a working fluid improves system performance significantly [9]. In 1970, significant research has been conducted on PVT hybrid systems [10]. The higher temperature raised the module's internal resistance and diminishes the power output. Wolf [11] has analyzed the flat-plate liquid-based PVT system for housing heating, based on the hourly electrical load. During the research of PVT collectors, Kern and Russell [12] considered air and water as heat removal fluids. Raghuraman [13] reported a numerical analysis of flat plate PVT collectors which use air/liquid for cooling purposes. Cox and Raghuraman [14] conducted computational models of air-cooled hybrid systems. The investigation by Sopian et al. [15] revealed that the effectiveness of a double-pass configuration is superior as compared to the single-pass design. Solanki et al. [16] explored indoor simulation and testing of photovoltaic/thermal (PVT) air collectors and predicted that the thermal and electrical effectiveness of the PVT solar heater is 42% and 8.4% at

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radiation and inlet temperatures of 600 W/m^2 and 38 °C, respectively.

The impact of glazing on the performance of photovoltaic thermal systems has been investigated by various authors. Glazing installed above the solar panel to capture additional heat energy almost doubles the thermal efficiency over the unglazed PVT, but a little reduction in electrical performance has been reported by Zondag [17]. Rajab et al. [18] examined the advantages of the anti-reflective layer in reducing reflection losses from top glass. Slimani et al. [19] perform an experimental study to assess the effects of glazing on the effectiveness of four distinct PVT designs. They accomplished that a DPPVT system with a glass cover exceeds the other designs in terms of thermal performance. Hegazy [20] investigated four distinct models by utilizing glazed PVT air systems. He observed that Model I had the poorest efficiency when compared to other models with nearly identical performance. However, Models IV and II require more fan power than Model III. Erzat et al. [21] conducted an experimental study to produce energy by utilizing the PV module with a solar thermal system. They found that the proposed hybrid technique generates 2.8 kWh of thermal energy each day. Teo et al. [22] conducted an experimental study to see the influence of active cooling on the hybrid PVT system and predicted that the maximum heat absorbed at the value of the mass flow rate of 0.055 kg/s. After exceeding this limit, the system has not been affected by the flow rate.

The utilization of PVT-heated air for different applications has been investigated by many researchers. Tiwari et al. [23] recommended the utilization of an air-type PVT design for heating the biogas. Three PV panels have 35W power and are enclosed in a glass conduit and the air is distributed by fans. The temperature of the heated air has been obtained from 38 to 47 °C to utilize the heat for the greenhouse. The parametric study based on different design configurations has been investigated by Tiwari et al. [24]. They noticed that as the module length increases, the overall efficiency significantly reduces due to increased system losses. The performance of a photovoltaic thermal system integrated with a thermoelectric generator module (PVT/TE) is numerically investigated by Ali Salari et al. [25] and compared with a PVT system. Based on the simulation results, the authors observed that the electrical efficiency of the PVT/TE system is much higher than that of the PVT system, while the PVT system benefits from higher thermal efficiency in comparison to the PVT/TE system. Alireza Akbar et al. [26] evaluated the performance of a thermoelectric-based PVT system for four distinct cooling fluids and compared the optimized and experimental results. Alfegi et al. [27] investigated the role of air-fluid velocity on the effectiveness of double-duct single-pass PVT design using fins. According to the findings, raising the mass flow rate increases total efficiency from 27.5 to 40.4%. Qader et al. [28] investigated the effectiveness of solar air heaters by utilizing a combined numerically and response surface methodology (RSM) technique. They recommended that the design parameters be optimized for efficient operation at the length of fin = 1.52 mm, slant angle = 49, Reynolds number = 18243.5, and pitch of fin = 19.04 mm.

According to the literature, the PVT system has different design configurations in which the double-pass produces superior performance due to the augmentation of heat, lowering the PV module's temperature. However, the significance of having a PVT system in a recycling operation associated with RSM has received little attention and study. Therefore, the main objectives of the current study are:

- To utilize the RSM (response surface methodology) technique to optimize the variables to provide an outstanding outcome for the current study.
- To develop the mathematical correlations to predict the thermal and electrical efficiencies.
- To establish the influence of recycling operations on PVT systems and to propagate pertinent data to developers, experts, and users.
- To recognize the most significant input factor influencing the thermal and electrical performance of the PVT system.

Furthermore, the RSM was used to investigate the interaction of four parameters, including recycling ratio, varying channel depth, air mass flow rate, and depth ratio, with the thermal and electrical performance of a double-pass hybrid PVT system. It is observed that the anticipated mathematical model associated with the RSM technique is significant with high precision.

2. Theoretical analysis

Fig. 1 depicts the current investigation, which considers the combined performance of an air-based PVT system under the recycling function. It consists of a photovoltaic module, a single glass cover, and the circulation of the air in the upper channel having its temperature corresponding to ambient temperature T_a. Air with mass flow rate $\dot{m}$ merges with air detached from the lower channel with $\dot{m}G$ before joining the upper channel. The equivalent climatic, operating, and PVT system parameters used are given in Table .1 to contrast the overall performance.

Based on the assumptions mentioned by authors [30], the energy balance equations for the various elements of the recycling type PVT system as shown in Fig. 1 have been presented below:

(a) Top glass:

$$I_g \alpha_g = h_{r(g-s)} (T_g - T_s) + h_{c(g-w)} (T_g - T_a) + h_{c(g-f1)} (T_g - T_{f1}) + h_{r(g-pv)} (T_g - T_{pv}) \quad (1)$$

(b) Air flowing through the upper channel

$$\frac{\dot{m}(1+G)c_p}{W} \frac{dT_{f1}}{dx} = h_{c(g-f1)} (T_g - T_{f1}) + h_{c(pv-f1)} (T_{pv} - T_{f1}) \quad (2)$$

(c) Photovoltaic cells

$$I_g \tau_g \alpha_{pv} (1-P) + I_g \tau_g \alpha_{pv} \left(1 - \eta_{pv}\right) = h_{c(pv-f1)} (T_p - T_{f1}) + h_{r(g-pv)} (T_{pv} - T_g) + h_{c(pv-f1)} (T_{pv} - T_{f2}) + h_{r(pv-bp)} (T_{pv} - T_{bp}) \quad (3)$$

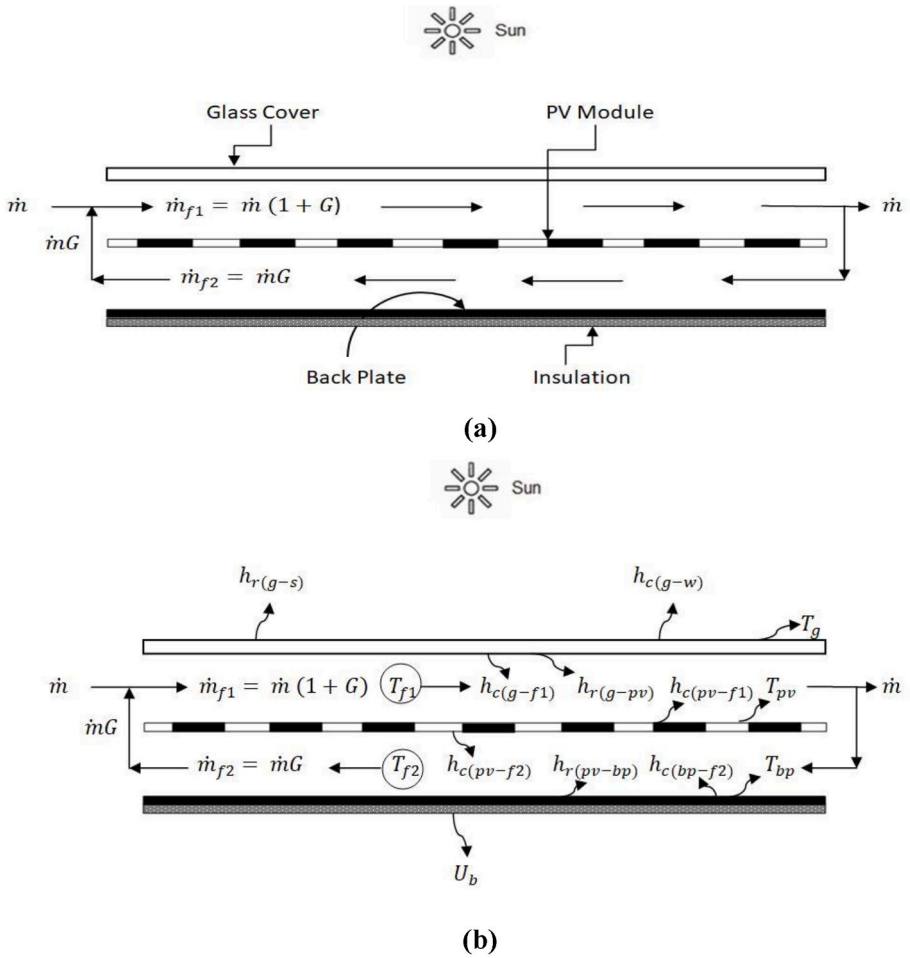


Fig. 1. Schematic diagram of recyclic type PVT design representing (a) the air flow pattern (b) the various heat transfer coefficients.

Table 1
The range of the parameters required for simulation [29].

Parameters	Values	Parameters	Values
Incident radiation (I)	300 - 1000 W/ m^2	Emissivity of backplate (ε_b)	0.7
Recycle ratio (G)	0.3–1.8	Emissivity of PV panel (ε_p)	0.7
Mass flow rate ($\dot{m}$)	0.03–0.15 kg/s	Efficiency of PV cells at standard test condition (η_{0p})	0.18
Depth ratio (D)	1- 6		
Packing factor (P)	0.40 - 0.95	Transmissivity of glass (τ_g)	0.92
Length of PV cell (L)	1 m	Absorptivity of collector plate (α_p)	0.9
Width of PV cell (W)	0.7 m	Initial or ambient Temperature	298 K
Emissivity of glass (ε_g)	0.85	PV module's absorptivity (α_{pv})	0.9
Wind velocity (u)	1 m/s	Absorptivity of glass (α_g)	0.04

(d) Air flowing through the lower channel

$$\frac{-\dot{m}Gc_p}{W}\frac{dT_{f2}}{dx} = h_{c\,(bp-f2)}\,\left(T_{bp}-T_{f2}\right) + h_{c\,(pv-f2)}\,\left(T_{pv}-T_{f2}\right) \tag{4}$$

(e) Back plate

$$h_{r\,(pv-bp)}\,\left(T_{pv}-T_{bp}\right) = U_b\left(T_{bp}-T_a\right) + h_{c\,(bp-f2)}\,\left(T_{bp}-T_{f2}\right) \tag{5}$$

The expressions for T_{pv}, T_g and T_{bp} could be written by utilizing the gauss elimination technique in Eqs. (1), (3) and (5) as follows:

$$T_{pv} = x_{11} + x_{12}T_{f1} + x_{13}\,T_s + x_{14}\,T_a + x_{15}T_{f2} \tag{6}$$

$$T_g = y_{11} + y_{12}T_{f1} + y_{13}\,T_s + y_{14}\,T_a + y_{15}T_{f2} \tag{7}$$

$$T_{bp} = z_{11} + z_{12}T_{f1} + z_{13}\,T_s + z_{14}\,T_a + z_{15}T_{f2} \tag{8}$$

Where the sky temperature [30] is given by

$$T_s = 0.552\,T_a^{1.5}$$

The two first-order linear differential equations are derived by simply replacing the values T_{pv}, T_g and T_{bp} in Eqs. (2) and (4) as given below:

$$\frac{dT_{f1}}{dx} - u_1\,T_{f1} - u_2T_{f2} = v_1 \tag{9}$$

$$\frac{dT_{f2}}{dx} - u_3T_{f1} - u_4T_{f2} = v_2 \tag{10}$$

Eqs. (9) and (10), subject to boundary conditions, constitute a two-point boundary value problem, which is normally solved by a numerical method. However, a closed-form solution can be obtained by the following procedures [31]. Firstly, Eqs. (10) and (11), can be combined as:

Eq. (9) + λ Eq. (10)

$$\frac{d}{dx}\left(T_{f_1}+\lambda T_{f_2}\right)=(u_1+\lambda u_3)T_{f_1}+(u_2+\lambda u_4)T_{f_2}+(v_1+\lambda v_2) \tag{11}$$

Rearrange the above equations, as given below:

$$\frac{d}{dx}\left(T_{f_1}+\lambda T_{f_2}\right)=u_1+\lambda u_3\left(T_{f_1}+\left(\frac{u_2+\lambda u_4}{u_1+\lambda u_3}\right)T_{f_2}\right)+(v_1+\lambda v_2) \tag{12}$$

This is further expressed as:

$$\frac{dy}{cy+k}=dx \tag{13}$$

Where,

$$c=u_1+\lambda u_3, k=v_1+\lambda v_2, \lambda=\frac{(u_2+\lambda u_4)}{u_1+\lambda u_3}, y=T_{f_1}+\lambda T_{f_2}$$

Where λ gives the quadric equation which has two roots as given below

$$\lambda=\frac{(u_4-u_1)\pm\sqrt{(u_1-u_4)^2+4u_2u_3}}{2u_3}$$

and,

$$k_1=v_1+\lambda_1 v_2, k_2=v_1+\lambda_2 v_2, c_1=u_1+\lambda_1 u_3, c_2=u_1+\lambda_2 u_3$$

The general solution of Eq. (13) can be written as [31]:

$$y=-\frac{k}{c}+\frac{Q}{c}e^{cx} \tag{14}$$

where Q is the arbitrary constant. Eq. (14) can also be written as

$$T_{f_1}+\lambda T_{f_2}=-\frac{k}{c}+\frac{Q}{c}e^{cx} \tag{15}$$

By putting the two roots in Eq. (15) which gives the yield of the temperatures of air as a function of x and can be written as [31]:

$$T_{f_1}(x)=\frac{1}{\lambda_2-\lambda_1}\left(\frac{k_2\lambda_1}{c_2}-\frac{k_1\lambda_2}{c_1}+\frac{Q_1\lambda_2e^{c_1x}}{c_1}-\frac{Q_2\lambda_1e^{c_2x}}{c_2}\right) \tag{16}$$

$$T_{f_2}(x)=\frac{1}{\lambda_2-\lambda_1}\left(-\frac{k_2}{c_2}+\frac{k_1}{c_1}-\frac{Q_1e^{c_1x}}{c_1}+\frac{Q_2e^{c_2x}}{c_2}\right) \tag{17}$$

2.1. Determination of the constants Q_1 and Q_2

The constants of Eqs. (16) and (17) can be evaluated by applying the appropriate boundary conditions as given below:

At the inlet of the system;

$$T_{f_1}|_{x=0}=\frac{G\times T_{f_2}|_{x=0}+T_a}{1+G} \tag{18}$$

Put the above boundary condition in Eq. (16) we get

$$-\frac{Q_1}{c_1}\left(\lambda_2+\frac{G}{1+G}\right)+\frac{Q_2}{c_2}\left(\lambda_1+\frac{G}{1+G}\right)=\frac{k_2}{c_2}\left(\lambda_1+\frac{G}{1+G}\right)-\frac{k_1}{c_1}\left(\lambda_2+\frac{G}{1+G}\right)-\left(\frac{\lambda_2-\lambda_1}{1+G}\right)T_a \tag{19}$$

At the outlet of the system;

$$T_{f_1}|_{x=L}=T_{f_2}|_{x=L}$$

$$-\frac{Q_1}{c_1}e^{c_1L}(1+\lambda_2)+\frac{Q_2}{c_2}e^{c_2L}(1+\lambda_1)=\frac{k_2}{c_2}(1+\lambda_1)-\frac{k_1}{c_1}(1+\lambda_2) \tag{20}$$

Eq. (19) and Eq. (20) can be written as.

AX = B.Where,

$$\mathrm{A}=\begin{bmatrix}-\left(\lambda_2+\frac{G}{1+G}\right) & \lambda_1+\frac{G}{1+G}\\ -e^{c_1L}(1+\lambda_2) & e^{c_2L}(1+\lambda_1)\end{bmatrix}, X=\begin{bmatrix}Q_1\\Q_2\end{bmatrix}$$

$$\mathrm{B}=\begin{bmatrix}\frac{k_2}{c_2}\left(\lambda_1+\frac{G}{1+G}\right)-\frac{k_1}{c_1}\left(\lambda_2+\frac{G}{1+G}\right)-\left(\frac{\lambda_2-\lambda_1}{1+G}\right)T_a\\ \frac{k_2}{c_2}(1+\lambda_1)-\frac{k_1}{c_1}(1+\lambda_2)\end{bmatrix}$$

The values of Q_1 and Q_2 obtained by solving the above matrix are helpful to calculate the temperature of flowing fluid in the upper and lower channel by substituting in Eq. (16) and Eq. (17), respectively. Correspondingly, the temperature of T_{pv}, T_g, T_{bp} are obtained by putting the values T_{f_1} and T_{f_2} in Eqs (6)–(8). The various heat transfer coefficients are considered by applying the procedure as discussed by the author [30].

The fan power ( ) needed to blow air into the channel, and is expressed as [32]:

$$P_{fan}=\frac{P_{flow}}{\eta_{fan}\eta_{motor}} \tag{21}$$

Here the motor efficiency (η_{motor}) and fan efficiency (η_{fan}) are presumed to be 70% and 90%, respectively. However the calculations of the mechanical power (P_{flow}) is different as given below [33]:

$$P_{flow}=\frac{\dot{m}(1+G)(\Delta P_1)+\dot{m}G(\Delta P_2)}{\rho} \tag{22}$$

Where ΔP_1 and ΔP_2 is the pressure drop across the upper and lower channel respectively, and can be calculated as:

$$\Delta P_1=4f_1\left(\frac{\rho v_1^2}{2}\right)\left(\frac{L}{d_{h1}}\right) \tag{23}$$

$$\Delta P_2=4f_2\left(\frac{\rho v_2^2}{2}\right)\left(\frac{L}{d_{h2}}\right) \tag{24}$$

Here f is the friction factor and the subscript 1 and 2 indicated the upper and lower channel and is given as:

For laminar flow [27]:

$$f_1=\frac{16}{Re_1} \tag{25}$$

and,

$$f_2=\frac{16}{Re_2} \tag{26}$$

For turbulent flow [27].

$$f_1=0.0591\,Re_1^{-0.2} \tag{27}$$

and

$$f_2=0.0591\,Re_2^{-0.2} \tag{28}$$

The net electrical power of the PV module is given as [34]:

$$P_{Net}=I_g\ \tau_g\ \alpha_{PV}\ \eta_{pv}\ P-P_{fan} \tag{29}$$

The accuracy of the developed mathematical design is authenticated via comparing the analytical outcomes with the experimental outcomes obtained by authors [34] and excellent agreement among the outcomes. Therefore, these analytical results have been used to identify the optimized parameter by utilizing RSM at which the PVT system gives superior performance. All the heat transfer coefficients and constant terms have been provided in the Appendix section.

3. Procedure for the iterative solution

To start the computer program which is developed by the author [30], the initial temperatures of all PVT system elements are considered to be the same as the ambient temperature. After that, the heat transfer coefficients were considered. To calculate the temperature of the elements, the Gauss elimination process is utilized to resolve a matrix produced by PVT system elements. After that, the temperatures of the PV cells are calculated by inserting the subsequently estimated temperatures and heat transfer coefficients into the governing equation. The process was repeated until the change in the temperature of the PV panel is less than 10^{-6}.

4. Response surface methodology (RSM)

RSM is a widely known statistical technique for designing experiments. RSM mainly consists of three primary steps: conducting statistically planned experiments, investigating the correlations for a mathematical model with response prediction, and assessing the model's competency. It is extremely beneficial for estimating the response or attention influenced through many input parameters to improve the responses. This technique has also established the associations among more than one calculated response and the vital controllable input parameters. It can choose the most appropriate range of experimental variables to generate a maximum or minimum significance level and could use 2D and 3D graphs to depict the straightforward and dynamic impact of the working variables.

When all factors are supposed to be computable, the parameters could be written as follows:

$$y = f(x_1, x_2, \ldots\ldots\ldots\ldots x_n) \tag{30}$$

Where y is the system's response and x_n are the variables of action known as factors. Using the relation [35], the overall number of testing executed to investigate the effect of four independent variables may be

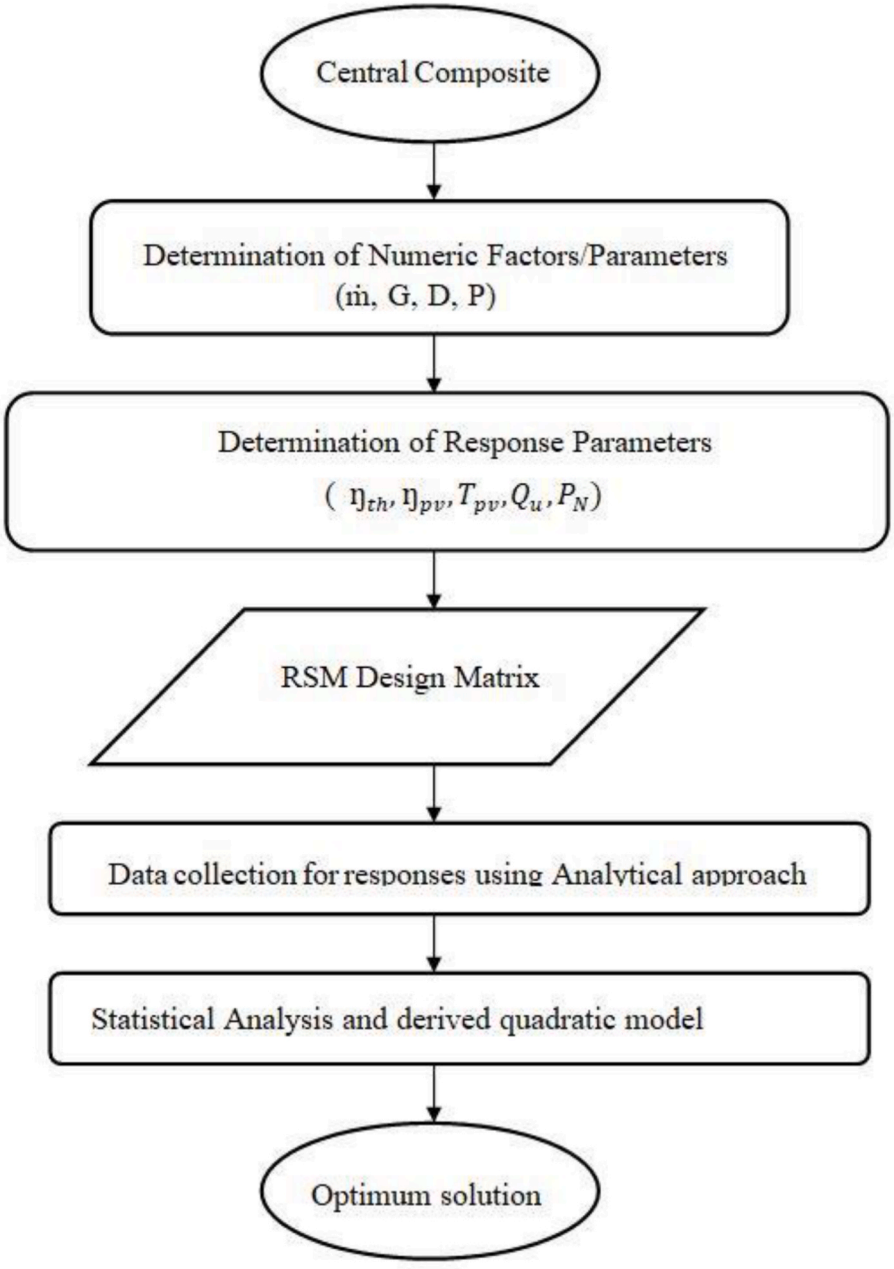


Fig. 2. RSM employment process flowchart for recycling type PVT system.

Table 2
Design of RSM and analytically computed outcomes (I = 800W/m^2).

Run	ṁ, (kg/s), (A)	G, (B)	D, (C)	P, (D)	η_{th}	η_{pv}	T_{pv} (°C)	Q_u (W)	P_N (W)
1	0.15	1.5	1.5	0.8	0.7189	0.155	55.8858	402.611	80.0523
2	0.09	1.5	3	0.7	0.7007	0.1533	57.9425	392.366	70.5652
3	0.15	1.8	6	0.5	0.9492	0.1642	44.4482	531.539	21.0345
4	0.15	1.8	1	0.8	0.6751	0.1531	58.2408	378.042	78.0265
5	0.09	1.5	3	0.7	0.7007	0.1533	57.9425	392.366	70.5652
6	0.15	0.3	1	0.5	0.4791	0.132	84.248	268.315	47.7123
7	0.09	1.5	3	0.7	0.7007	0.1533	57.9425	392.366	70.5652
8	0.09	0.3	3	0.8	0.4647	0.135	80.5634	260.25	75.3632
9	0.15	0.3	6	0.95	0.5949	0.1493	62.9106	333.15	90.9712
10	0.15	1.8	6	0.95	0.8925	0.1655	42.95	499.791	68.9759
11	0.03	1.8	6	0.5	0.5513	0.1405	73.8163	308.72	44.2689
12	0.15	0.3	6	0.5	0.6372	0.147	65.7421	356.815	45.5446
13	0.09	1.8	1.5	0.5	0.654	0.1477	64.9306	366.24	47.2336
14	0.09	1.5	1.5	0.7	0.6036	0.1464	66.4252	338.03	64.0884
15	0.03	0.3	1	0.95	0.2113	0.1077	114.218	118.344	65.6441
16	0.15	0.3	1	0.95	0.4479	0.1349	80.6605	250.806	92.0222
17	0.03	1.8	1	0.95	0.3562	0.1249	93.0418	199.475	85.6933
18	0.15	1.8	1	0.95	0.6592	0.1538	57.3849	369.154	91.3825
19	0.03	1.8	6	0.95	0.5158	0.143	70.6418	288.855	86.0521
20	0.09	1.5	3	0.7	0.7007	0.1533	57.9425	392.366	70.5652
21	0.09	1.5	3	0.7	0.7007	0.1533	57.9425	392.366	70.5652
22	0.09	1.5	3	0.8	0.6902	0.1538	57.3948	386.526	79.0861
23	0.03	0.3	6	0.5	0.3324	0.1173	102.437	186.118	44.8323
24	0.03	1.8	1	0.5	0.3781	0.1218	96.8877	211.737	45.3843
25	0.09	1.5	3	0.7	0.7007	0.1533	57.9425	392.366	70.5652
26	0.03	0.3	1	0.5	0.2216	0.1045	118.14	124.108	30.5558
27	0.03	1.5	3	0.8	0.4595	0.1349	80.6396	257.318	75.2882
28	0.09	1.5	1.5	0.7	0.6036	0.1464	66.4252	338.03	70.9825
29	0.03	0.3	6	0.95	0.3143	0.1204	98.5397	175.996	81.8365
30	0.09	1.5	3	0.7	0.7007	0.1533	57.9425	392.366	70.5652

computed as:

$$N = 2^n + 2n + n_c = 2^4 + 2 \times 4 + 6 = 30 \quad (31)$$

The central composite design (CCD) is the most commonly used response surface design for optimizing the model. It is made up of 2^k factorial design points, which are supplemented by multiple center points and 2^k axial (star) points. In the present study, central composite designs have five levels for each variable, however, this can be changed by selecting a = 1.0, a face-centered CCD. Each variable in the face-centered design has only four levels. The 2^k exploratory factorial points were utilized to adjust a quick command, and the longitudinal direction runs were then further to allocate the quadratic elements to be integrated into the design. Fig. 2 depicts a schematic representation of the RSM technique. The schematic depicts the procedures necessary to construct the RSM-developed model and the optimal solution for the input variables. The values of the G, ṁ, D, and P were used as process variables for this investigation. Once all the process variables have been identified, consequently, the results of the responses are set. In the current study, five response factors (η_{th}, $\eta_{pv}, T_{pv}, Q_u, P_N$) were taken into account, and the resulting RSM design matrix for the current device is displayed in Table 2. To achieve the best result, the responses have been attached in a quadratic model by utilizing regression analysis. On the

Table 3
ANOVA table for thermal efficiency.

Source	Sum of Squares	Degree of freedom	Mean Square	F-value	p-value	
Model	0.9393	14	0.0671	1380.96	<0.0001	significant
A-Mass flow rate	0.4001	1	0.4001	8235.44	<0.0001	
B-Recycle ratio, G	0.2231	1	0.2231	4591.92	<0.0001	
C-Depth ratio, D	0.1028	1	0.1028	2116.77	<0.0001	
D-Packing factor, P	0.0051	1	0.0051	104.91	<0.0001	
AB	0.0080	1	0.0080	164.81	<0.0001	
AC	0.0022	1	0.0022	45.90	<0.0001	
AD	0.0007	1	0.0007	13.47	0.0023	
BC	0.0040	1	0.0040	83.35	<0.0001	
BD	0.0004	1	0.0004	8.95	0.0091	
CD	0.0000	1	0.0000	0.3199	0.5800	
A^2	0.0053	1	0.0053	109.54	<0.0001	
B^2	0.0029	1	0.0029	60.55	<0.0001	
C^2	0.0024	1	0.0024	50.24	<0.0001	
D^2	0.0006	1	0.0006	11.44	0.0041	
Residual	0.0007	15	0.0000			
Lack of Fit	0.0007	8	0.0001			
Pure Error	0.0000	7	0.0000			
Cor Total	0.9401	29				

other hand, RSM has the potential to optimize and forecast the thermal and electrical performance for the current study.

5. Mathematical designing and the RSM design matrix

RSM is employed in the present study to develop a statistical relationship among the process variables and their response based on the results of the analytical study. As described in Table 2, Design @ expert software is utilized to create a design matrix, and the required output has been estimated with the help of an analytical technique that uses MATLAB code. RSM, one of the most widely used statistical modeling approaches for analyzing experimental or mathematical statistics with the least amount of work was adopted to examine the research findings.

5.1. RSM design

From this approach, a data matrix is developed and derived with a four-factor face-centered CCD with complete recombination. The G, $\dot{m}$, D, and P are the key process variables examined for this investigation. After that from the recombination of the data (Table 2) the regression analysis, as well as the correlation and outcomes are derived. The capability of this modeling technique is assessed by utilizing the chronological F-test and the statistics (ANOVA) approach in a similar package to achieve the most excellent models. The step-wise regression approach is utilized, which automatically eliminates inconsequential design terms during the analysis of variance (ANOVA). Tables 3 and 4 exhibit the data of ANOVA for thermal and electrical efficiencies, respectively. By using sequential F-tests the $\eta_{th}, \eta_{pv}, T_{pv}, Q_u, and\ P_N$ suggest a quadratic connection, where the auxiliary expressions are important, and the designs are not aliased. Moreover, the ANOVA results show that the $\dot{m}$ is the most important variable that influences the thermal and electrical effectiveness of the double-pass PVT system, as demonstrated by F values.

Tables 5 and 6 describe the findings of the measure of variability in observed data of thermal and electrical efficiency in terms of R^2 and Adj. R^2. In the ANOVA table, the value of the significant coefficient R^2 is illustrated as the ratio of variance described by the model to the whole variance. The value of R^2 closer to one means the superior response model fits the actual data [36]. The predicted R^2 values for thermal and electrical efficiency presented in Tables 5 and 6 respectively are satisfactorily close to unity. Furthermore, the value of R^2 shows that this model accounts for almost 99% of the variability in data. As a result, it is established that this model contributes a good description of the association among the independent factors and the responses for both thermal and electrical efficiency.

Table 5
Measure of variability in observed responses of thermal efficiency.

Std. Dev.	0.0070	R^2	0.9992
Mean	0.5772	**Adjusted R^2**	0.9985
C.V. %	1.21	**Predicted R^2**	0.9949
		Adeq Precision	150.6184

Table 6
Measure of variability in observed responses of electrical efficiency.

Std. Dev.	0.0012	R^2	0.9972
Mean	0.1424	**Adjusted R^2**	0.9945
C.V. %	0.8232	**Predicted R^2**	0.9718
		Adeq Precision	76.5212

The following are the final mathematical correlations established by the Design @ Experts Software 13:

$$\eta_{th} = 0.162 + 4.903\,A + 0.378 \times B + 0.078 \times C - 0.739 \times D + 0.513 \times A \times B + 0.081 \times A \times C - 0.508 \times A \times D + 0.009 \times B \times C - 0.032 \times B \times D - 0.001 \times C \times D - 15.0 \times A^2 - 0.133 \times B^2 - 0.008 \times C^2 + 0.513 \times D^2 \quad (32)$$

$$\eta_{pv} = 0.068 + 0.569 \times A + 0.032 \times B + 0.007 \times C + 0.010 \times D - 0.012 \times A \times B - 0.004 \times A \times C - 0.012 \times A \times D + 0.0001 \times B \times C - 0.0009 \times B \times D - 0.0002 \times C \times D - 1.714 \times A^2 - 0.008 \times B^2 - 0.0006 \times C^2 - 0.001 \times D^2 \quad (33)$$

$$T_{pv} = 163.6 - 702.9 \times A - 39.3 \times B - 9.21 \times C - 13.36 \times D + 14.52 \times A \times B + 4.44865 \times A \times C + 14.82 \times A \times D - 0.211 \times B \times C + 1.19 \times B \times D + 0.274 \times C \times D + 2117.7 \times A^2 + 10.51 \times B^2 + 0.756 \times C^2 + 1.20 \times D^2 \quad (34)$$

Table 4
ANOVA table for electrical efficiency.

Source	Sum of Squares	Degree of freedom	Mean Square	F-value	p-value	
Model	0.0072	14	0.0005	376.66	<0.0001	significant
A-Mass flow rate	0.0030	1	0.0030	2152.45	<0.0001	
B-Recycle ratio, G	0.0016	1	0.0016	1178.27	<0.0001	
C-Depth ratio, D	0.0008	1	0.0008	587.25	<0.0001	
D-Packing factor, P	0.0000	1	0.0000	17.42	0.0008	
AB	4.291E-06	1	4.291E-06	3.12	0.0976	
AC	4.515E-06	1	4.515E-06	3.29	0.0899	
AD	3.014E-07	1	3.014E-07	0.2193	0.6463	
BC	1.488E-06	1	1.488E-06	1.08	0.3145	
BD	3.749E-07	1	3.749E-07	0.2728	0.6091	
CD	2.420E-07	1	2.420E-07	0.1761	0.6807	
A^2	0.0001	1	0.0001	50.02	<0.0001	
B^2	0.0000	1	0.0000	8.56	0.0104	
C^2	0.0000	1	0.0000	9.00	0.0090	
D^2	2.618E-09	1	2.618E-09	0.0019	0.9658	
Residual	0.0000	15	1.374E-06			
Lack of Fit	0.0000	8	2.577E-06			
Pure Error	0.0000	7	0.0000			
Cor Total	0.0073	29				

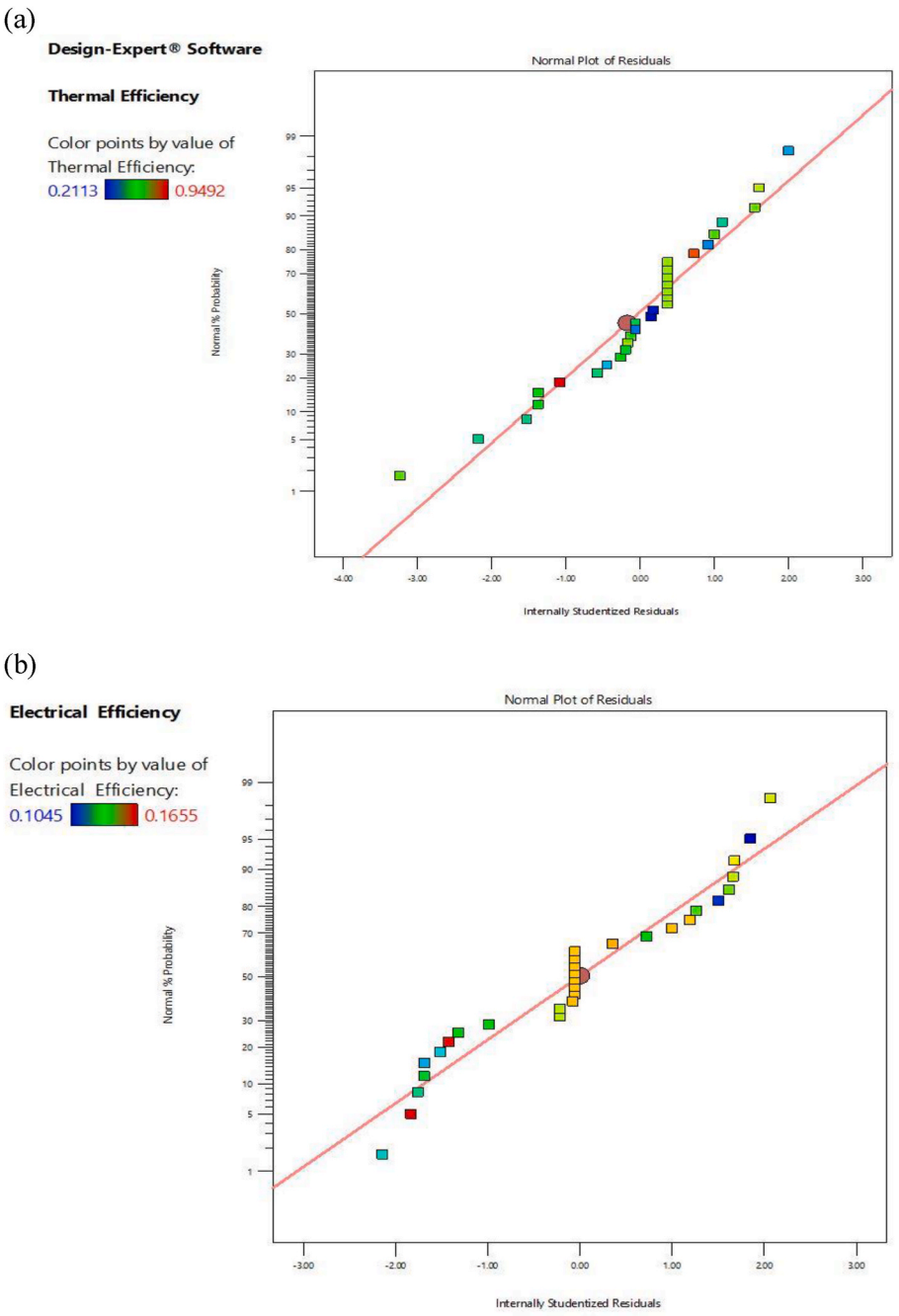


Fig. 3. Normal probability plots for (a) thermal efficiency (b) electrical efficiency of the recycling type PVT system.

$$Q_u = 90.57 + 2745.81 \times \mathrm{A} + 211.28 \times \mathrm{B} + 43.89 \times \mathrm{C} - 413.95 \times \mathrm{D} + 286.79 \times \mathrm{A} \times \mathrm{B} + 45.39 \times \mathrm{A} \times \mathrm{C} - 284.48 \times \mathrm{A} \times \mathrm{D} + 4.79 \times \mathrm{B} \times \mathrm{C} - 17.97 \times \mathrm{B} \times \mathrm{D} - 1.02 \times \mathrm{C} \times \mathrm{D} - 8448.5 \times \mathrm{A}^2 - 74.43 \times \mathrm{B}^2 - 4.9 \times \mathrm{C}^2 + 287.01 \times \mathrm{D}^2 \quad (35)$$

$$P_N = -46.2249 + 257.843 \times A + 24.164 \times B + 10.6763 \times C + 117.175 \times D - 120.711 \times A \times B - 33.3431 \times A \times C + 152.118 \times A \times D - 2.44432 \times B \times C + 6.15618 \times B \times D + 0.433745 \times C \times D - 570.505 \times A^2 - 4.80296 \times B^2 - 0.838207 \times C^2 - 30.8004 \times D^2 \quad (36)$$

5.2. Validation

The normal probability graphs of the residuals for thermal and electrical efficiency are exposed in Fig. 3(a) and (b), respectively. The residuals are often lying in a straight line, indicating that the errors are fairly allocated. It also suggests that the established regression statistical method can produce appropriate results.

To authenticate the established RSM models, the results for thermal and electrical efficiency from the analytical technique are contrasted with the RSM results, as displayed in Fig. 4 (a) and 4(b), respectively. According to the validation process, the estimated error between the observed and forecast values is low, indicating satisfactory conformity between the created RSM and the analytical model. As a result, the models are utilized to accurately estimate the output values of a recycling PVT system.

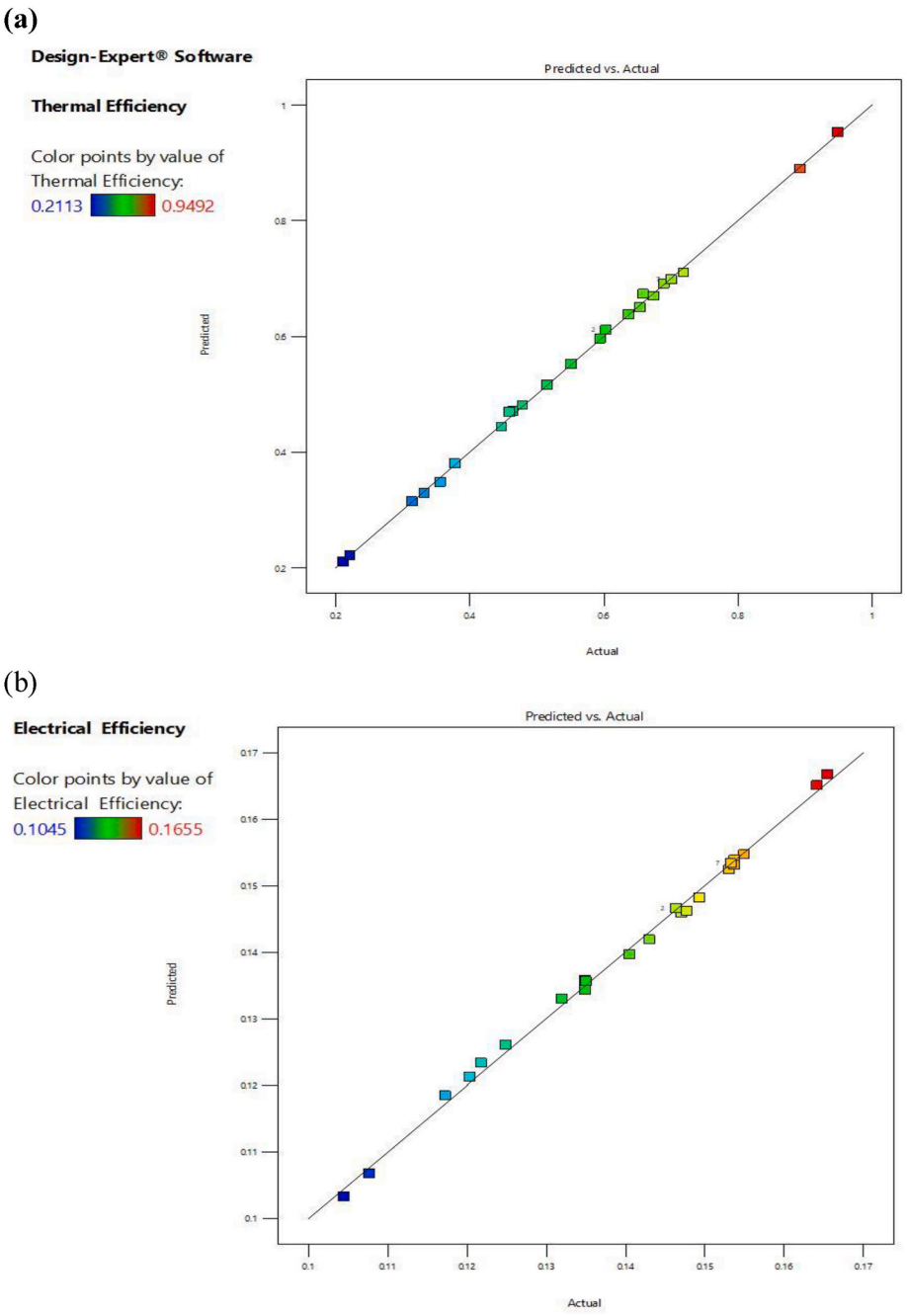


Fig. 4. The confirmation of the established RSM designs with the analytical (a) thermal efficiency (b) electrical efficiency.

6. Results and discussion

The effectiveness of the RSM model in stipulations of the thermal and electrical performance of the PVT system under external recycle operation was studied by utilizing an analytical and RSM as a combined method. The major purpose of employing RSM is to identify the system's optimum operating conditions and an area that satisfies the terms and conditions. An analytical model was developed in the present research by utilizing MATLAB to demonstrate an iterative resolution technique. The RSM is then employed to organize the arrangements of processing variables and evaluate their responses of concern via constructing a statistical equation using simulation results. As indicated in Table 1, the range of the processing variables (ṁ, G, D, and P factor) used for thermal and electrical prediction are 0.3–1.8, 0.03–0.15 (kg/s), 1–6, and 0.5–0.95, respectively.

6.1. Influence of G, ṁ, D, and P on thermal and electrical efficiency

The perturbation plot in Fig. 5 demonstrated the influence of G, ṁ, D, and P on thermal efficiency. Basically, in the perturbation plot, all the factors are considered for one response. It has been revealed that increasing the values of the G, ṁ, and D raises the thermal efficiency while it decreases with increasing the value of the P of the proposed PVT system as shown in Fig. 6. This is because the recycling phenomenon increases the convection process i.e., coefficient of heat transfer because of higher velocity at a given value of ṁ and preheats the flowing fluid at the channel's entrance, resulting in a considerable increase in heat transfer rate. Furthermore, the recycling ratio ṁG creates the bulk motion in the double pass PVT system's upper channel, and the heat exclusion rate is directly proportional to the value of ṁ as shown in Fig. 6(a).

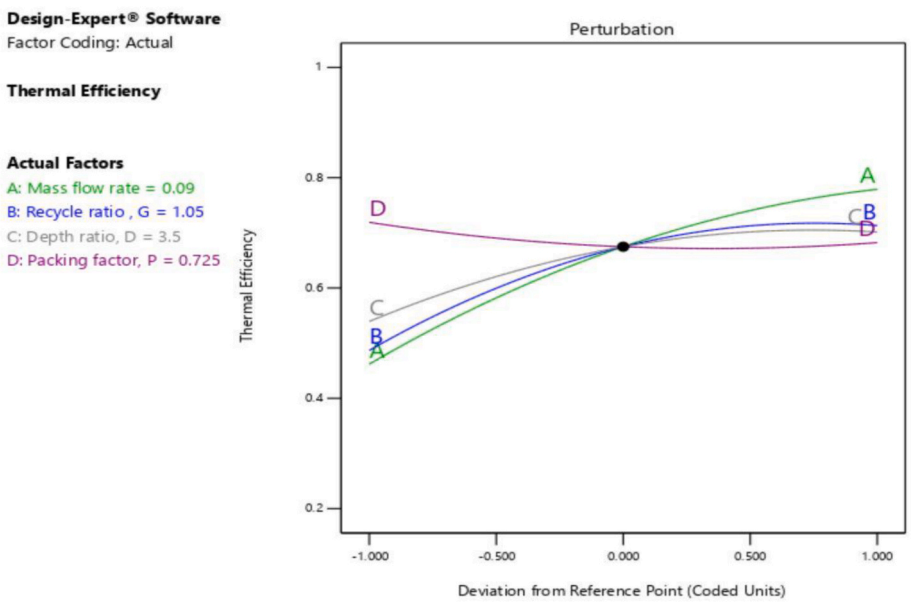


Fig. 5. Variation of the thermal efficiency of recycling type PVT system with ṁ (A), G (B), D (C), and P (D).

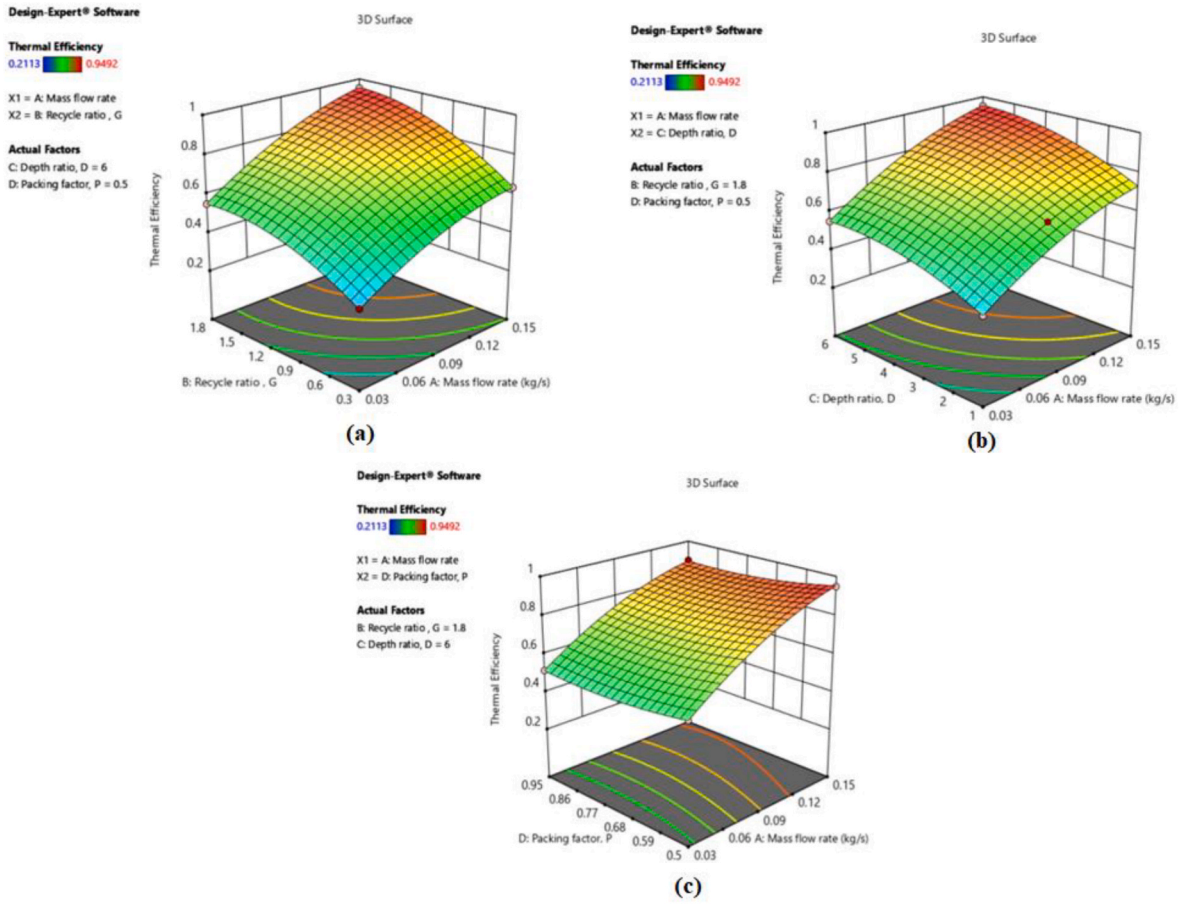


Fig. 6. Effect of (a) G and ṁ (b) ṁ and D (c) ṁ and P on the thermal efficiency of recyclic type PVT system.

Whereas it is considered that all portions of the outer part of the PVT system, are well thermally insulated excluding the glass cover. Furthermore, an increase in the depth ratio means diminishing the depth of the lower channel due to which the speed of the air increases and consequential growth in the thermal efficiency as shown in Fig. 6(b). The packing factor is a further crucial constraint that has drastically affected the performance of the PVT system behind the value of D. As we amplify the values of the P then electrical performance increases while thermal performance decreases as shown in Fig. 6(c). As indicated in Table 2, the maximum value of thermal efficiency is approximately

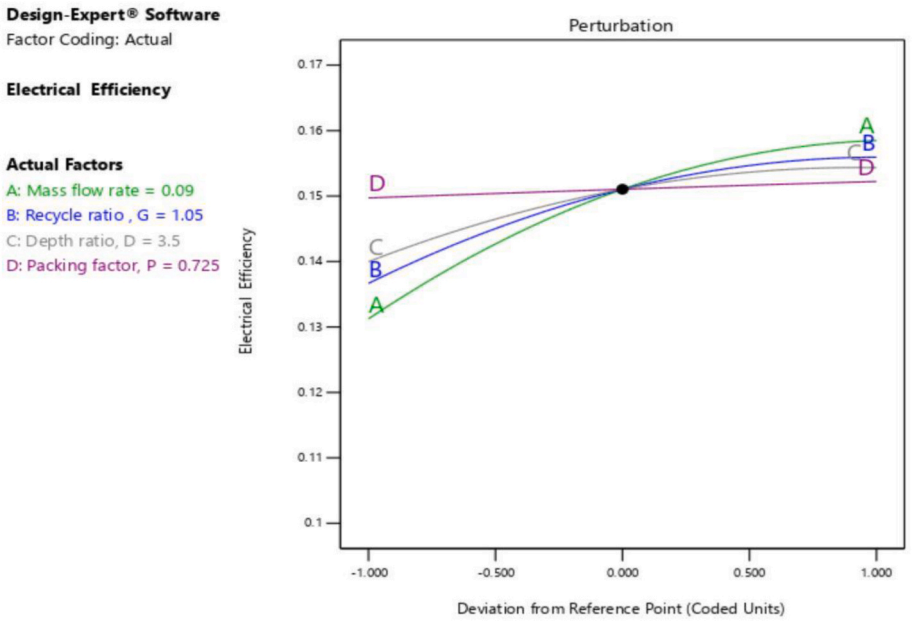


Fig. 7. Variation of the electrical efficiency of recycling type PVT system with ṁ (A), G (B), D (C), and P (D).

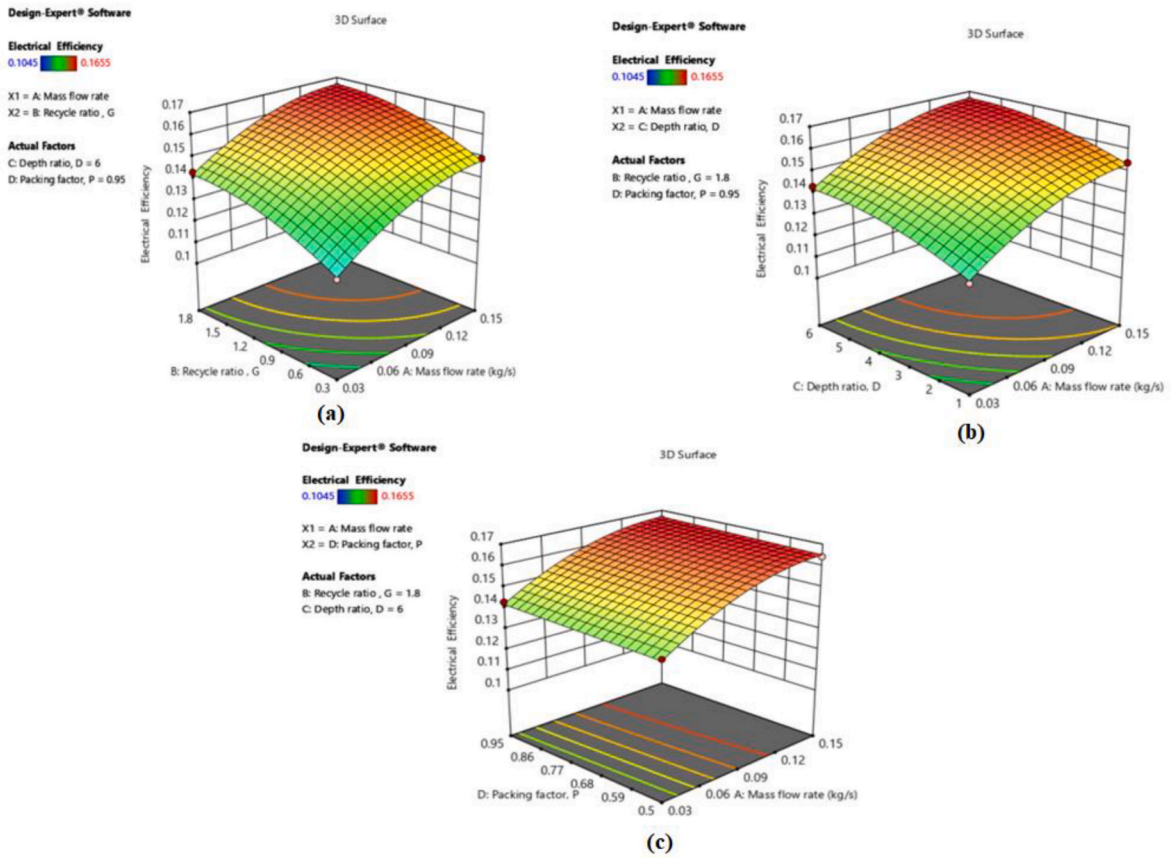


Fig. 8. Effect of (a) G and ṁ (b) ṁ and D (c) ṁ and P on the electrical efficiency of recyclic type PVT system.

94.92% when G = 1.8, ṁ = 0.15 kg/s, D = 6, and P = 0.5.

Figs. 7 and 8 depict the variation of the electrical efficiency of the PVT system with the values of ṁ, G, D, and P. It is clear in Fig. 8, the electrical efficiency of the PVT system increases with all the factors. As indicated in Table 2, the maximum value of electrical efficiency is approximately 16.55% when G = 1.8, ṁ = 0.15 kg/s, D = 6, and P = 0.95.

6.2. *Influence of ṁ and G on PV cell's temperature*

The influence of G and ṁ on the temperature of the PV cell is presented in Fig. 9(a). The results demonstrate that as the values of G and ṁ

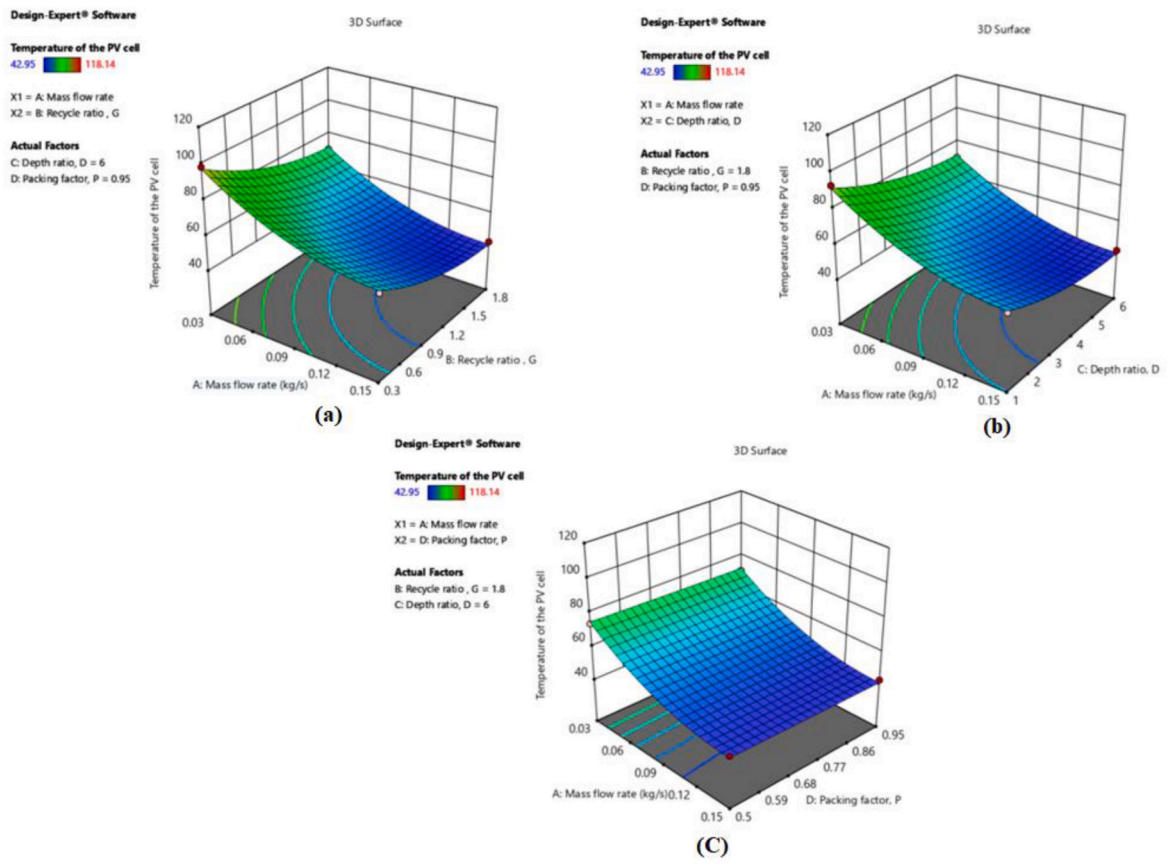


Fig. 9. Influence on the PV cell's temperature of the (a) ṁ and G (b) ṁ and D (c) ṁ and P.

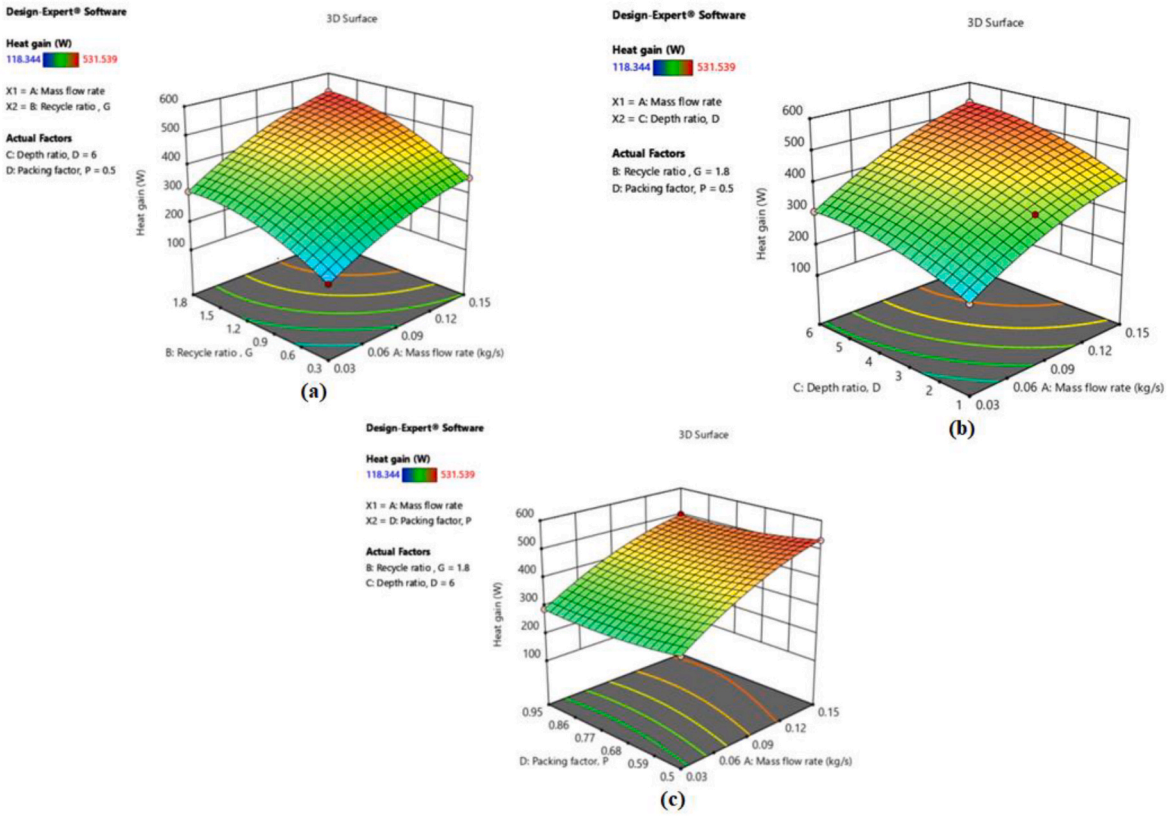


Fig. 10. Influence on the heat gain of the (a) ṁ and G (b) ṁ and D (c) ṁ and P.

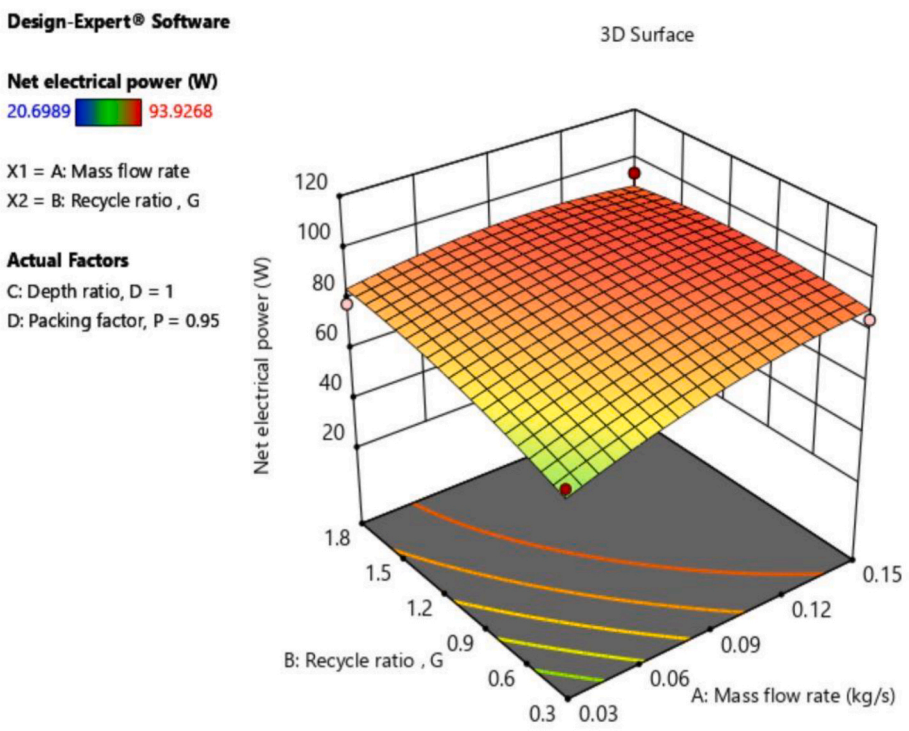


Fig. 11. Influence of the ṁ and G on P_N.

increase the temperature of the PV cell drops. The value of G raises the heat transfer coefficients, while the ṁ raises the heat extraction rate, resulting in a decrease in PV cell temperature. The temperature of the PV cell drops as the channel depth ratio (D) increases, as seen in Fig. 9(b). It has been observed that increasing the depth ratio decreases the lower channel depth due to which the turbulence rate increases resulting in a decrement in the PV panel temperature of the PVT system. As seen in Fig. 9(c), raising the packing factor reduces the temperature of the PV panel. At the higher value of the packing factor, the additional exterior part of the PV module is accountable to produce electrical energy due decrement in the PV cell temperature.

6.3. Influence of ṁ, G, D, and P on the useful heat gain and net electrical power

Fig. 10 (a) depicts the variation of useful heat gain of the present PVT system with the values of G and ṁ at the fixed value of the D and P. An equivalent effect of the values of G, ṁ, D, and P on the performance of the present PVT system is shown in Fig. 10(b) and (c). The heat gain increases with the increase in the value of G and ṁ because of an increase in heat transfer coefficients. It is also found that the gain of heat increases significantly with the increase in the depth ratio and decreases with the packing factor. Fig. 11 represents that raising the values of ṁ and G simultaneously reduces the system's power due to the rising pressure drop.

6.4. Optimization

The purpose of this approach is to identify several constraints that will satisfy all the targets. The desirability value does not have to be 1.0 because it is entirely contingent on how close the higher and lower limit is placed to the exact optimum value. Design @ Expert13 statistical software is used to generate a range of 30 perfect solutions. The combination of variables with the uppermost close to one is considered the optimum state for essential outcomes. Once the optimum value of the operating parameters has been established, the final phase is to estimate

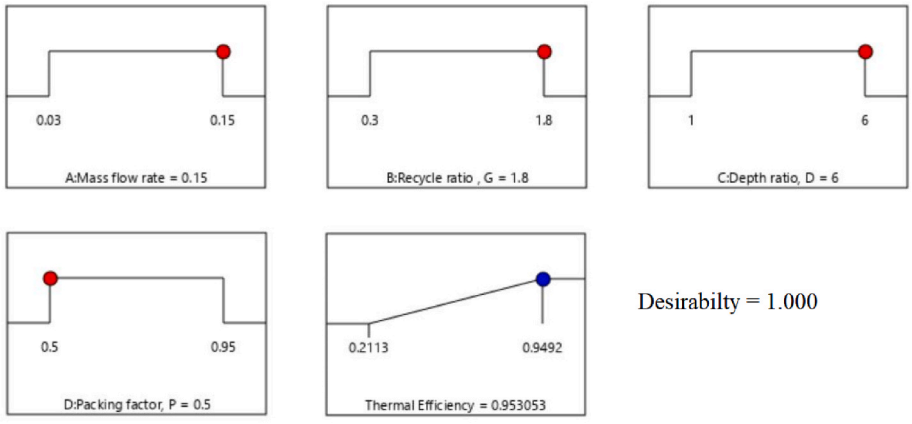


Fig. 12. Ramp function diagram of desirability for thermal efficiency.

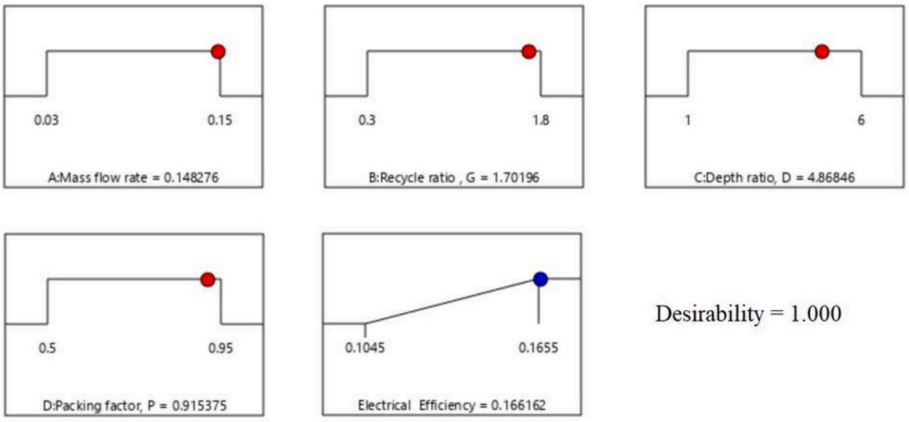


Fig. 13. Ramp function diagram of desirability for electrical efficiency.

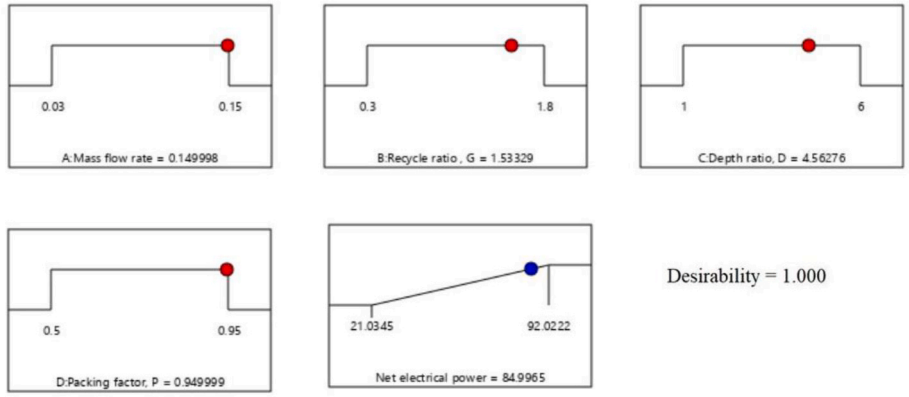


Fig. 14. Ramp function diagram of desirability for net electrical power of the PVT system.

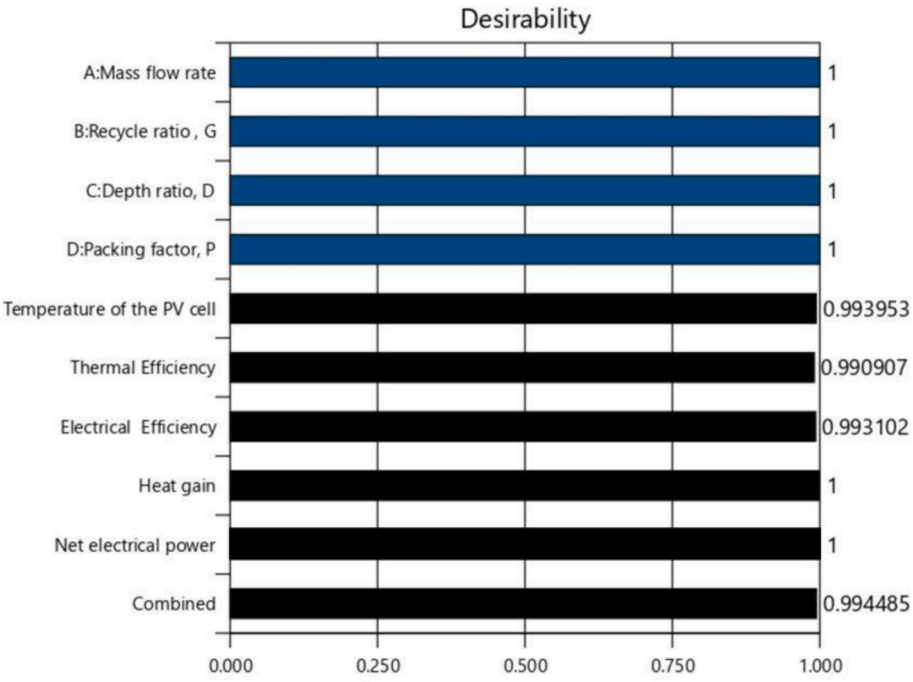


Fig. 15. Bar diagram of desirability for all responses.

Table 7
Comparison of the analytical result and RSM result at the optimum values for net electrical power.

Optimum design Parameter	P_N (W)		
	Analytical value	RSM value	Error (%)
$\dot{m}$ = 0.14 kg/s, G = 1.53, D = 4.56, P = 0.95	85.35	85.00	0.42

and confirm the enhancement of the performance requirements using the optimum condition of the PVT system's parameters. To test the feasibility of yield responses, the values of $\dot{m}$, G, D, and P were all adjusted to be within range, while thermal and electrical efficiencies were chosen to be maximum.

Figs. 12–14 illustrate the ramp function graphs of desirability for thermal, electrical efficiency, and net electrical power respectively, while Fig. 15 shows the bar graphs of desirability for all responses. According to Figs. 12–14, the best result for thermal efficiency is achieved at the value of $\dot{m}$ = 0.15 kg/s, G = 1.8, D = 6, and P = 0.5, for electrical efficiency is gained at the value of $\dot{m}$ = 0.1483, G = 1.702, D = 4.87, and P = 0.915 and for net electrical power is achieved at the value of $\dot{m}$ = 0.14, G = 1.53, D = 4.56, and P = 0.95. The total desirability function of the responses is represented by a bar graph. Desirability ranges from 0 to 1 based on how close the result is to the goal. The bar graph depicts how well each variable meets the criterion values i.e., near one is considered satisfactory.

In Table 7, the result of the RSM model is compared with the numerical study at optimum values of the parameters. The validation results demonstrate that the deviation is very small, with only a 0.41% difference among the result, which indicates positive conformity among the RSM and the analytical design.

7. Conclusion

A PVT system under recycling mode was used in this study to determine the most suitable design variables. The RSM with the center composite design (CCD) was used to optimize the design specifications. The main findings of the current design are mentioned below:

1. According to the literature review, various modifications have been suggested and implemented to improve the cooling rate. However, the significance of having a PVT system in a recycling operation associated with RSM has received little attention and study. As a result, in this study, this design was proposed.
2. ANOVA findings demonstrate that mass flow rate is the most significant input factor influencing the thermal and electrical performance of the PVT system, followed by channel depth ratio, recycle ratio, and packing factor.
3. It was discovered that the quadratic design established by the RSM to estimate various responses and all responses applied with a high degree of accuracy to assess the performance of the PVT system. The model was subjected to a comprehensive statistical evaluation to ensure its validity and precision.
4. The result of the RSM model is compared with the analytical study at the most favorable values of the parameters. The validation results demonstrate that the deviation is very small, with only a 0.41% difference among the result, which indicates positive conformity among the RSM and the analytical design. The validation part confirmed the final goal of the study.
5. The optimized values of the design factors for the efficient performance of the current design to supply the optimal net electrical power of 85W were established to be at $\dot{m}$ = 0.14 kg/s, G = 1.53, D = 4.56, and P = 0.95. These optimized values correspond to the desirability of approximately exhibiting a tremendously desirable response. As a result, it is concluded that the methodology designed in this research work to improve the cooling of the PVT system is effective and utilizes the appropriate analytical techniques.
6. This study's main investigation focuses on how recycling affects a photovoltaic thermal system's performance. Techno-economic analyses, however, are left out of the investigation and can be carried out in subsequent studies. To view the velocity and temperature profile diagram, it is also necessary to conduct a CFD investigation of the proposed recycling type PVT system for various airflow patterns.

CRediT authorship contribution statement

Ajay Kumar: Conceptualization, Methodology, Software, Writing – original draft. **Prashant Dhiman:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

$$x_{11=}\frac{L_2L_6+L_8}{1-LL_6-L_{11}L_7}, x_{12}=\frac{L_3L_6+L_9}{1-L_1L_6-L_{11}L_7},\ x_{13}=\frac{L_4L_6}{1-L_1L_6-L_{11}L_7}$$

$$x_{14}=\frac{L_5L_6+L_7L_{12}}{1-L_1L_6-L_{11}L_7}, x_{15}=\frac{L_{10}+L_7L_{13}}{1-L_1L_6-L_{11}L_7}, y_{11}=\frac{L_2+L_8L_1-L_2L_7L_{11}}{1-L_7L_{11}-L_1L_6}$$

$$y_{12}=\frac{L_3+L_1L_9-L_7L_{11}L_3}{1-L_7L_{11}-L_1L_6}, y_{13}=\frac{L_4-L_4L_7L_{11}}{1-L_7L_{11}-L_1L_6}$$

$$y_{14}=\frac{L_5+L_1L_7L_{12}-L_5L_7L_{11}}{1-L_7L_{11}-L_1L_6}, y_{15}=\frac{L_1L_{10}+L_1L_7L_{13}}{1-L_7L_{11}-L_1L_6}$$

$$z_{11}=\frac{L_{11}L_2L_6+L_{11}L_8}{1-L_1L_6-L_{11}L_7}, z_{12}=\frac{L_{11}L_9+L_3L_{11}L_{13}+L_{13}-L_{13}L_1L_6-L_{11}L_7L_{13}}{1-L_1L_6-L_{11}L_7}$$

$$z_{13}=\frac{L_{11}L_4L_6}{1-L_1L_6-L_{11}L_7}, z_{14}=\frac{L_{11}L_5L_6+L_{11}L_7L_{12}+L_{12}-L_{12}L_1L_6-L_{12}L_{11}L_7}{1-L_1L_6-L_{11}L_7}$$

$$z_{15}=\frac{L_{11}L_{10}+L_{11}L_7L_{13}}{1-L_1L_6-L_{11}L_7}, L_1=\frac{h_{r\ (g-p)}}{h_{r\ (g-s)}+h_{c\ (g-w)}+h_{c\ (g-f1)}+h_{r\ (g-p)}}$$

$$L_2 = \frac{I_g \alpha_g}{h_{r\ (g-s)} + h_{c\ (g-w)} + h_{c\ (g-f1)} + h_{r\ (g-p)}}$$

$$L_3 = \frac{h_{c\ (g-f1)}}{h_{r\ (g-s)} + h_{c\ (g-w)} + h_{c\ (g-f1)} + h_{r\ (g-p)}}$$

$$L_4 = \frac{h_{r\ (g-s)}}{h_{r\ (g-s)} + h_{c\ (g-w)} + h_{c\ (g-f1)} + h_{r\ (g-p)}}$$

$$L_5 = \frac{h_{c(g-w)}}{h_{r\ (g-s)} + h_{c\ (g-w)} + h_{c\ (g-f1)} + h_{r\ (g-p)}}$$

$$L_6 = \frac{h_{r\ (g-p)}}{h_{c\ (p-f1)} + h_{r\ (g-p)} + h_{c\ (p-f2)} + h_{r\ (p-bp)}}$$

$$L_7 = \frac{hr}{h_{c\ (p-f1)} + h_{r\ (g-p)} + h_{c\ (p-f2)} + h_{r\ (p-bp)}}$$

$$L_7 = \frac{h_{r\ (p-bp)}}{h_{c\ (p-f1)} + h_{r\ (g-p)} + h_{c\ (p-f2)} + h_{r\ (p-bp)}}$$

$$L_8 = \frac{I_g \tau_g \alpha_g (1-p) + I_g \tau_g \alpha_{pv} \left(1-\eta_{pv}\right) p}{h_{c(p-f1)} + h_{r(g-p)} + h_{c(p-f2)} + h_{r(p-bp)}}$$

$$L_9 = \frac{hc}{h_{c\ (p-f1)} + h_{r\ (g-p)} + h_{c\ (p-f2)} + h_{r\ (p-bp)}}$$

$$L_{10} = \frac{hc}{h_{c\ (p-f1)} + h_{r\ (g-p)} + h_{c\ (p-f2)} + h_{r\ (p-bp)}}$$

$$L_{11} = \frac{h_{r\ (p-bp)}}{U_b + h_{c\ (bp-f2)} + h_{r\ (p-bp)}}, L_{12} = \frac{U_b}{U_b + h_{c\ (bp-f2)} + h_{r\ (p-bp)}}$$

$$L_{13} = \frac{h_{c(bp-f2)}}{U_b + h_{c\ (bp-f2)} + h_{r\ (p-bp)}}$$

$$u_1 = -\left(s_3 - s_2 x_{12} - s_1 y_{12}\right), u_2 = s_2 x_{15} + s_1 y_{15}), u_3 = -\left(r_1 z_{12} + r_2 x_{12}\right)$$

$$u_4 = r_3 - r_1 z_{15} - r_2 x_{15})$$

$$v_1 = s_2 x_{11} + s_1 y_{11} + s_2 x_{13} T_s + s_2 x_{14} T_a + s_1 y_{13} T_s + s_1 y_{14} T_a$$

$$v_2 = -r_1 z_{11} - r_1 z_{13} T_s - r_1 z_{14} T_a - r_2 x_{11} - r_2 x_{13} T_s - r_2 x_{14} T_a$$

$$s_1 = \frac{h_{c(g-f1)} w}{\dot{m}(1+G) c_p}, s_2 = \frac{h_{c(p-f1)} w}{\dot{m}(1+G) c_p}, s_3 = \frac{\left(h_{c(g-f1)} + h_{c(p-f1)}\right) w}{\dot{m}(1+G) c_p},$$

$$r_1 = \frac{h_{c(bp-f2)} w}{\dot{m}(1+G) c_p}, r_2 = \frac{h_{c(p-f2)} w}{\dot{m}(1+G) c_p}, r_3 = \frac{\left(h_{c(bp-f2)} + h_{c(p-f2)}\right) w}{\dot{m}(1+G) c_p}$$

Nomenclature

ŋ	Efficiency $ŋ_{th}$ Thermal efficiency
$ŋ_{op}$	Nominal efficiency A_c Area of the collector plate
η_{pvt}	Photovoltaic thermal efficiency c_P Specific heat of air
σ	Stefan's Boltzmann constant G Recycle ratio
h_r	Radiation heat transfer coefficient I Incident radiation
h_c	Convection heat transfer coefficient D Depth ratio
k_f	The conductivity of the fluid (air) L Length of collector plate
$\dot{m}$	Mass flow rate of air P_N Net electrical power
T	Temperature P Packing factor
Re_1	Reynolds number for upper channel T_{ref} Reference temperature
Re_2	Reynolds number for lower channel W Width of PV panel

Subscript

a	Ambient air bp Back plate

f1 Air at upper channel f2 Air at the lower channel
g Glass g − s Among glass and sky
g − w Among glass and wind pv Photovoltaic (PV) cells
pv − f1 Among PV cells and air at the upper channel
g − pv Among glass and PV cells
pv − f2 Among PV cells and air at the lower channel
pv − bp Among PV cells and back plate
bp − *f*2 Among the back plate and air at the lower channel

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2022.12.053.

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